SHUBHAM KUMAR

Software Engineer Lead

Azure Cloud | Generative AI | Java & Python

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EXPERIENCE

Software Engineer Lead

Sep 2026 - Present

Optum Global Solutions (UHG) — Noida, India

- Lead the development of AI-enabled engineering solutions using LLMs, RAG, Python, FastAPI, and Microsoft Azure for incident analysis and operational efficiency.
- Develop an incident-intelligence solution that analyzes ServiceNow incidents and retrieves relevant technical and operational context to support investigation and resolution.
- Design and deploy Azure workloads with Functions, App Service, Storage, Key Vault, managed identity, Application Insights, and Log Analytics.
- Build secure cloud infrastructure with VNets, subnets, private endpoints, firewall rules, RBAC, Terraform, GitHub Actions, and Azure OIDC.
- Lead architecture discussions, code reviews, cloud modernization, and engineering practices across application, platform, SRE, and security teams.

Technologies: Azure, LLMs, RAG, Python, FastAPI, ServiceNow, Azure Functions, App Service, Terraform, GitHub Actions, Key Vault, Managed Identity

Senior Software Engineer

Mar 2025 - Aug 2026

Optum Global Solutions (UHG) — Noida, India

- Integrated CodeQL and Prisma Cloud scanning into Jenkins and GitHub Actions pipelines to strengthen vulnerability detection and secure delivery.
- Designed conversational and knowledge-retrieval services using Python, FastAPI, LLM APIs, vector databases, and RAG workflows.
- Led code reviews and design discussions for Spring Boot microservices and React applications, improving maintainability, security, and engineering quality.
- Improved production reliability and incident analysis using Splunk, Kibana, New Relic, and Grafana while mentoring engineers and supporting cross-functional delivery.

Technologies: Java, Spring Boot, React.js, Python, FastAPI, RAG, AWS, Docker, Kubernetes, Jenkins, GitHub Actions, CodeQL, Prisma Cloud

Software Engineer

Jul 2022 - Feb 2025

Optum Global Solutions (UHG) — Noida, India

- Developed enterprise digital-security applications using Java, Spring Boot, React.js, REST APIs, Kafka, and MongoDB.
- Built APIs for authorization, organization-access management, provider registration, and digital-security workflows supporting systems used by 1.4M+ users.
- Resolved 200+ high-priority ServiceNow incidents, supporting application stability and operational continuity.
- Performed MongoDB migrations, indexing, and performance optimization and developed automated tests with JUnit, Mockito, and JMeter.

Technologies: Java, Spring Boot, React.js, MongoDB, Kafka, REST APIs, ServiceNow, JWT, Docker, JUnit, Mockito, JMeter

EDUCATION

Master of Technology (M.Tech) - Information Security

Atal Bihari Vajpayee Indian Institute of Information Technology and Management

Aug 2020 - May 2022 | Gwalior, India

Bachelor of Technology (B.Tech) - Computer Science

West Bengal University of Technology

Aug 2015 - Jul 2019 | Kolkata, India

SUMMARY

Software Engineer Lead with 4+ years of experience building enterprise applications and cloud solutions across Microsoft Azure, generative AI, Java, and Python. Builds LLM/RAG applications with Python and FastAPI, and scalable services with Java, Spring Boot, React.js, Kafka, and MongoDB. Experience includes Azure Functions, App Service, Key Vault, private networking, Terraform, CI/CD, observability, secure SDLC, production support, architecture reviews, and technical leadership. Track record includes resolving 200+ high-priority ServiceNow incidents and contributing to digital-security workflows serving 1.4M+ users.

SKILLS

Languages
Java, Python, JavaScript, C++, SQL

Generative AI
LLMs, RAG, AI Agents, Prompt Engineering, Semantic Search, OpenAI APIs, LangChain, Vector Databases

Application
Spring Boot, FastAPI, REST APIs, Microservices, Kafka, React.js, Redux

Azure / Cloud
Azure Functions, App Service, Storage, Key Vault, Managed Identity, RBAC, Application Insights, Log Analytics, VNets, Subnets, Private Endpoints, AWS

DevOps / Security
Terraform, Docker, Kubernetes, GitHub Actions, Jenkins, Azure OIDC, CI/CD, Spring Security, JWT, OWASP, CodeQL, Prisma Cloud, HashiCorp Vault

Data / Observability / Testing
MongoDB, MySQL, Splunk, Kibana, New Relic, Grafana, JUnit, Mockito, JMeter

ADDITIONAL EXPERIENCE

Teaching Assistant

Atal Bihari Vajpayee Indian Institute of Information Technology and Management, Gwalior

Jan 2022 - May 2022

- Supported C Programming and Artificial Intelligence labs, conducted live coding sessions, and assisted students with programming assignments and technical problem-solving.

SELECTED AI & ENTERPRISE PROJECTS

LLM-Powered Incident Intelligence & Resolution Assistant

- Develop an AI-assisted solution that analyzes ServiceNow incidents with LLM/RAG workflows and surfaces relevant operational context for troubleshooting and resolution.
- Build Python/FastAPI services and deploy Azure Functions and App Service components with Terraform-based private networking and secure identity.

Technologies: LLMs, RAG, Python, FastAPI, ServiceNow, Azure Functions, App Service, Terraform, Private Networking

AI Digital Security Chatbot

- Built a FastAPI assistant that indexes engineering repositories and answers security and engineering questions using context-aware retrieval.
- Applied RAG and LLM workflows to improve the discovery of technical knowledge across engineering repositories.

Technologies: Python, FastAPI, RAG, LLMs, AI Agents, Semantic Search

AI Allure Report Agent

- Built an AI agent with FastAPI to analyze Allure test reports and generate actionable summaries for stakeholders after test execution.

Technologies: Python, FastAPI, AI Agents, Automation, Allure

Digital Security for Provider Portal

- Supported authorization workflows serving 1.4M+ users and high-volume transactions; developed APIs for authorization, organization access, and TIN registration.
- Implemented secure access controls using Spring Security, JWT, and HashiCorp Vault.

Technologies: Java, Spring Boot, React.js, MongoDB, Kafka, REST APIs, Spring Security, JWT

Enhanced Provider Validation (EPV) Portal

- Developed provider-registration and validation workflows using Spring Boot, React.js, REST APIs, MongoDB, document submission, and Kafka-based tracking.

Technologies: Java, Spring Boot, React.js, MongoDB, Kafka, REST APIs, Pega

CERTIFICATIONS

Microsoft Certified: Azure Administrator Associate
Microsoft Learn — LINK

AWS Certified Cloud Practitioner
Amazon Web Services — LINK

Programming Foundations: Secure Coding
LinkedIn Learning — LINK

React - The Complete Guide 2024
Udemy — LINK

Core Java - Full Course
Udemy — LINK

AWS S3 Basics
Coursera — LINK

Blockchain Basics
Coursera — LINK

Optum AI Dojo Certification
Optum — LINK

PUBLICATION & ACHIEVEMENTS

Publication: Securing IoT with Blockchain: Detecting Malicious Nodes with TLS Certificates - IC2E3, 2023 | IEEE Xplore

GATE: Qualified Computer Science in 2019, 2020, and 2022.

CodeChef: Achieved Global Rank 148 in the September Challenge 2021, Division 3.

HackerRank: Earned a 5-Star rating in C++ Programming.